Benchmark: **MA.7.DP.1.5** · Choose an appropriate graph Student code: _____ Calculator: yes

Learning goal: I can choose and create an appropriate data display. (circle your level after scoring)

4	I can explain why one display is better than another for a given question.
3	I can choose the right display for numerical or categorical data and create it accurately.
2	I can make a bar graph or dot plot from data.
1	I need help to start this benchmark.

1. A student recorded the 280 kilograms of material collected at a recycling drive.

Paper	Plastic	Glass	Metal
126	70	56	28

The student wants a display that answers this question at a glance: is Paper more than half of the total? Which display is the most appropriate?

A. a bar chart
B. a histogram
C. a line plot
D. a circle graph

2. A class surveyed 140 students about the lunch they like best.
Pizza: 21
Tacos: 35
Salad: 63
Pasta: 21

The class wants a display that answers this question at a glance: did about one-fourth of the students choose Pizza? Which display is the most appropriate?

A. a histogram
B. a box plot
C. a circle graph
D. a bar chart

3. A gardener recorded the heights of 120 sunflowers, measured to the nearest centimeter. Almost every value is different. The table summarizes the data.

Height (cm)	Number
90–109	11
110–129	23
130–149	45
150–169	37
170–189	4

The gardener wants a display that shows the shape of the distribution. Which display is the most appropriate for this data set?

A. a circle graph
B. a histogram
C. a line plot
D. a stem-and-leaf plot

Turn over →

Benchmark: **MA.7.DP.1.5** · Choose an appropriate graph Student code: _____

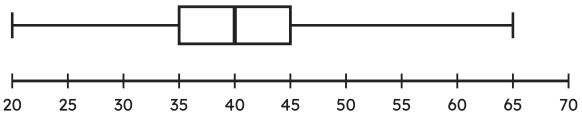
4. The data shows the number of pets for 16 students:
1, 0, 3, 1, 1, 2, 1, 0, 1, 2, 0, 1, 0, 0, 1, 2

Choose the display that shows every value and how often it occurs. Explain why it fits these data, and describe what it would look like.

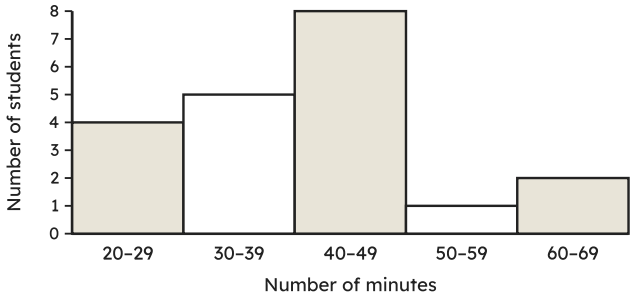
5. The two displays show the same data: the numbers of minutes 20 students spent reading over the weekend.

Which display can be used to find the interquartile range of the data? Explain why the other display cannot, and then use the display to find the interquartile range of the data.

Display A



Display B



Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-DP.1.5 · MA.7.DP.1.5 · Choose an appropriate graph

Unit 13: Numerical and Categorical Data. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **D. a circle graph**

The question compares one category with the whole. In a circle graph each section is the same fraction of the circle as its category is of the total, so a section can be compared with half of the circle at a glance (Paper is 45%).

A bar chart shows the amounts but not how each compares with the total; a histogram and a line plot are for numerical data.

2. Regular **C. a circle graph**

The question compares one category with the whole class. In a circle graph each section is the same fraction of the circle as its category is of the class, so half, one-third or one-fourth of the circle can be seen at a glance.

A bar chart shows the counts, but not how a count compares with the total; a box plot and a histogram are for numerical data.

3. Regular **B. a histogram**

There are 120 values, almost all different, so a line plot or a stem-and-leaf plot would need a mark or a leaf for nearly every value, and the table gives only counts for intervals.

The data are numerical, so a circle graph (parts of a whole) does not fit. A histogram uses the equal intervals in the table as its bars and shows the shape of the distribution.

4. Regular **A line plot, because the data are numbers with only a few different values, so every value can be shown. It would have an X above each value for each time it appears: 0 → 5 Xs, 1 → 7 Xs, 2 → 3 Xs, 3 → 1 X. Score: 1 point — a line plot or dot plot (a bar graph of the counts also accepted) with a reason that mentions the few different values, plus the description; the name alone earns 0.**

The data are numbers with only a few different values $(0,\ 1,\ 2,\ 3)$, so a line plot keeps every value and shows the shape; a stem-and-leaf plot would put every value on one stem, and a histogram would hide the individual values.

Count how many times each value occurs, then stack that many Xs above the value on a number line.

0: 5; 1: 7; 2: 3; 3: 1.

5. Rigor **Display A, the box plot, because it marks the median and the quartiles; the histogram groups the values into intervals, so it shows only which interval holds the quartiles, not the exact value. From it, the interquartile range is $45 - 35 = 10$. Score: 1 point — the display named, why the other cannot be used, and the value; the value alone earns 0.**

A box plot marks the minimum, Q_1, the median, Q_3 and the maximum, so the interquartile range is $Q_3 - Q_1 = 45 - 35 = 10$.

A histogram groups the values into intervals, so it shows only which interval holds the quartiles, not the exact value.

Re-check strip (3 items, C-DP.1.5-R)

1. Regular **D. a bar chart**

The question compares two counts. A bar chart shows one bar for each category, with heights read on a scale, so the two counts and their difference can be read.

A circle graph shows shares of the whole, not the counts; a box plot and a histogram are for numerical data.

2. Regular **A line plot, because the data are numbers with only a few different values, so every value can be shown. It would have an X above each value for each time it appears: 0 → 4 Xs, 1 → 3 Xs, 2 → 3 Xs, 3 → 2 Xs, 4 → 1 X. Score: 1 point — a line plot or dot plot (a bar graph of the counts also accepted) with a reason that mentions the few different values, plus the description; the name alone earns 0.**

The data are numbers with only a few different values $(0,\ 1,\ 2,\ 3,\ 4)$, so a line plot keeps every value and shows the shape; a stem-and-leaf plot would put every value on one stem, and a histogram would hide the individual values.

Count how many times each value occurs, then stack that many Xs above the value on a number line.

0: 4; 1: 3; 2: 3; 3: 2; 4: 1.

3. Regular **A. a stem-and-leaf plot**

The scores are numerical. A stem-and-leaf plot keeps every value (each leaf is one score), and the lengths of its leaf rows show the shape.

A box plot shows the shape and spread but not every value.

Comparison strip (2 items, C-DP.1.5-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **B. a bar chart**

The question compares two counts. A bar chart shows one bar for each category, with heights read on a scale, so the two counts and their difference can be read.

A circle graph shows shares of the whole, not the counts; a box plot and a histogram are for numerical data.

2. Regular **A. a circle graph**

The question compares one category with the whole. In a circle graph each section is the same fraction of the circle as its category is of the total, so a section can be compared with half of the circle at a glance (Rent is 45%).

A bar chart shows the amounts but not how each compares with the total; a histogram and a line plot are for numerical data.

Reteach plan — C-DP.1.5

MA.7.DP.1.5 · Choose an appropriate graph

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-DP.1.5-R

Benchmark: **MA.7.DP.1.5** · Choose an appropriate graph Student code: _____ Calculator: yes

1. A class surveyed 260 students about the way they get to school.
Bus: 52
Car: 39
Walk: 104
Bike: 26
Train: 39

The class wants a display that answers this question at a glance: how many more students chose Bus than chose Train? Which display is the most appropriate?

A. a histogram
B. a circle graph
C. a box plot
D. a bar chart

2. The data shows the number of pets for 13 students:
0, 4, 0, 2, 2, 1, 1, 3, 1, 0, 3, 2, 0

Choose the display that shows every value and how often it occurs. Explain why it fits these data, and describe what it would look like.

3. A teacher recorded the test scores of 15 students:
97, 97, 79, 95, 97, 90, 79, 87, 89, 72, 72, 72, 91, 85, 98

The teacher wants a display that keeps every value and shows the shape of the data. Which display is the most appropriate for this data set?

A. a stem-and-leaf plot
B. a box plot
C. a bar chart of categories
D. a circle graph

Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-DP.1.5-X

Benchmark: **MA.7.DP.1.5** · Choose an appropriate graph Student code: _____ Calculator: yes

1. A class surveyed 200 students about the way they get to school.
Bus: 60
Car: 20
Walk: 40
Bike: 50
Train: 30

The class wants a display that answers this question at a glance: how many more students chose Bus than chose Car? Which display is the most appropriate?

A. a box plot
B. a bar chart
C. a circle graph
D. a histogram

2. A student recorded how a family divides its monthly income of $4,800.

Rent	Food	Transportation	Savings
$2,160	$1,200	$960	$480

The student wants a display that answers this question at a glance: is Rent more than half of the total? Which display is the most appropriate?

A. a circle graph
B. a line plot
C. a histogram
D. a bar chart

Score: ____ / 2 → copy to your progress strip